

The Printer Working Group

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DRAFT IPP Specification

IPP Firmware Update Extensions v1.0 (FWUPDATE)

Status: Stable

Abstract: This specification defines new IPP attributes, values, and operations to enable a Client to request a Printer to check for new Firmware, report information about discovered new Firmware, and to install the available new Firmware.

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1. Introduction

Printer vendors provide new Firmware for Printers, and Printers support various mechanisms to install new Firmware. IPP System Service v1.1 [PWG 5100.22] provides a way to push new Firmware through the System. Conversely, some Printers are designed and implemented to autonomously contact a Firmware Repository and automatically update themselves with available new Firmware.

This specification defines new IPP attributes, values, and operations to enable a Client to request a Printer to determine whether there is available new Firmware, report information about available new Firmware, and request the Printer to install available new Firmware.

2. Terminology

2.1 Conformance Terminology

Capitalized terms, such as MUST, MUST NOT, RECOMMENDED, REQUIRED, SHOULD, SHOULD NOT, MAY, and OPTIONAL, have special meaning relating to conformance as defined in Key words for use in RFCs to Indicate Requirement Levels [BCP14]. This specification defines the following additional capitalized conformance terms:

CONDITIONALLY REQUIRED: A MUST conformance requirement that applies only when a specified condition is true.

DEPRECATED: A SHOULD NOT conformance requirement for previously defined and approved protocol elements that are planned to be removed from use.

OBSOLETE: A MUST NOT conformance requirement for previously defined and approved protocol elements that have been removed from use.

2.2 Printing Terminology

The following printing terms are used in this document:

Administrator: An End User who is also authorized to manage all aspects of a Printer, including creating the Printer instances and controlling the authorization of other End Users and Operators [STD92].

Authenticated User: The same as "authenticated user" defined in [STD92] RFC 8011 section 9.3.

End User: An End User is a person or software process that is authorized to perform basic printing functions, including finding/locating a Printer, creating a local instance of a Printer, viewing Printer status, viewing Printer capabilities, submitting a Print Job, viewing Print Job status, and altering the attributes of a Print Job [STD92].

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Logical Device: a print server, software service, or gateway that processes jobs and either forwards or stores the processed Job or uses one or more Physical Devices to render output [STD92].

Operator: An Operator is an End User that also has special rights on the Printer. The Operator typically monitors the status of the Printer and also manages and controls the Jobs at the Printer [STD92]. The Operator is allowed to query and control the Printer, Jobs, and Documents based on site policy.

Printer: Listener for incoming IPP session requests and receiver of incoming IPP operation requests (HTTP/1.1 server, as defined in [RFC7230]) that represents one or more Physical Devices or a Logical Device. [STD92].

Physical Device: a hardware implementation of an endpoint device, e.g., a marking engine, a fax modem, etc. [STD92]

2.3 Protocol Role Terminology

The following protocol roles are defined to specify unambiguous conformance requirements:

Client: Initiator of outgoing connections and sender of outgoing operation requests (Hypertext Transfer Protocol -- HTTP/1.1 [STD99] User Agent) [STD92].

Printer: Listener for incoming connections and receiver of incoming operation requests (Hypertext Transfer Protocol -- HTTP/1.1 [STD99]Server) that represents one or more Physical Devices or a Logical Device [STD92].

2.4 Other Terminology

This specification defines or adopts the following terms:

Acquisition: The new Firmware installation phase when the available new Firmware is downloaded from the Firmware Repository to the Printer.

Activation: The new Firmware installation phase when the installed new Firmware is activated for use.

Cleanup: The new Firmware installation phase when new firmware installation content is no longer needed and is deleted from the Printer.

Discovery: The new Firmware installation phase when available new Firmware is found by the Printer in the configured Firmware Repository.

Installation: The new Firmware installation phase when the acquired new Firmware is installed to make it available for Activation.

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Recovery: The new Firmware installation phase when the existing or previously installed Firmware is restored for use following an error with one of the other phases (Acquisition, Validation, Installation, Activation).

Validation: The new Firmware installation phase when the integrity of downloaded new Firmware is confirmed by the Printer.

Firmware: Persistent computer instructions and data embedded in the Printer that provides the basic functions of that device. Firmware is only replaced during a specialized update process. [PWG 5110.1]

Firmware Repository: A service that provides access to available Firmware packages and their metadata.

Firmware Update Agent: A software entity that interacts with a Firmware Repository.

Firmware Update Manager: A software entity, running locally or externally, that manages the process of installing new Firmware in a device.

Firmware Update: The act of replacing existing Firmware on a device such as a Printer. This includes the Installation, Validation, Activation and Cleanup phases, may include the Recovery phase if there are problems, and may include the Acquisition phase if that has yet to be performed.

Management Device: A device or set of devices that is used to manage Printers.

2.5 Acronyms and Organizations

This specification defines the following acronyms and organizations:

IANA: Internet Assigned Numbers Authority, https://www.iana.org/

IETF: Internet Engineering Task Force, https://www.ietf.org/

ISO: International Organization for Standardization, https://www.iso.org/

PWG: Printer Working Group, https://www.pwg.org/

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3. Requirements

3.1 Rationale

Given the following existing specifications:

1. IPP Driver Replacement Extensions v2.0 [PWG 5100.13] that defines the "printer-firmware-name", "printer-firmware-patches", "printer-firmware-string-version" and "printer-firmware-version" attributes to describe the Printer's current Firmware; and
2. IPP System Service v1.1 [PWG 5100.22] that defines operations and attributes that support managing and maintaining a Printer's resources, including its Firmware.

And given the existence of a wider range of Firmware update use cases as described in section 3.2, this IPP Firmware Update Extensions v1.0 specification should:

1. Define new operations to cause the Printer to check for new Firmware and install new Firmware;
2. Define new attributes to describe available new Firmware the Printer found by querying its Firmware Repository;
3. Define new values for indicating the Printer's state while it is installing new Firmware; and
4. Register all of these with IANA.

3.2 Use Cases

3.2.1 Check for New Firmware

Rafael has a document on his phone and chooses to print it. While he is looking at his print options, his phone indicates that new firmware is available for his printer. It offers to remind him later. He approves of being reminded later and prints his document.

3.2.2 Install New Firmware Immediately

Rafael is using his phone when it presents him with a notification informing him that new firmware is available for his printer, to see if he wishes to approve the update to occur. It gives him the option to install the new firmware immediately, install the new firmware at a specified later time, or not install the new firmware. Rafael chooses to install the new firmware immediately. The printer installs the new firmware immediately.

3.2.3 Install New Firmware Later

Rafael is using his phone when it presents him with a notification informing him that new firmware is available for his printer, to see if he wishes to approve the update to occur. It gives him the option to install the new firmware immediately, install the new firmware at a

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specified later time, or not install the new firmware. Rafael chooses to install the new firmware at 11:30pm that evening, then gets back to his email. He goes to sleep at 10:30pm.

At 11:30pm that evening, the printer installs the new firmware as configured earlier.

3.3 Exceptions

The following subsections define exceptions in addition to those defined in the Internet Printing Protocol/1.1 [STD92].

3.3.1 Update Firmware Successful

Rafael wakes up in the morning, and his phone presents a notification that his printer attempted to install new firmware at the chosen time, and that the new firmware was installed successfully.

3.3.2 Update Firmware Unsuccessful

Rafael wakes up in the morning, and his phone presents a notification that his printer attempted to install new firmware at the chosen time, but the update process failed. The phone presents a notification with a link to an assistance page.

3.4 Out of Scope

The following are considered out of scope for this specification:

1. Definition of Firmware file formats;
2. Supporting updating Printer with Firmware from an arbitrary URI; and
3. Special states a Printer is required to be in to receive and/or install available new Firmware.

3.5 Design Requirements

The design requirements for this specification are:

1. Define attributes and values to describe new Firmware discovered from the configured Firmware Repository;
2. Define operations for requesting the Printer to install available new Firmware;
3. Define security and privacy requirements necessary to support these operations and attributes; and
4. Define sections to register all attributes, values, and operations with IANA.

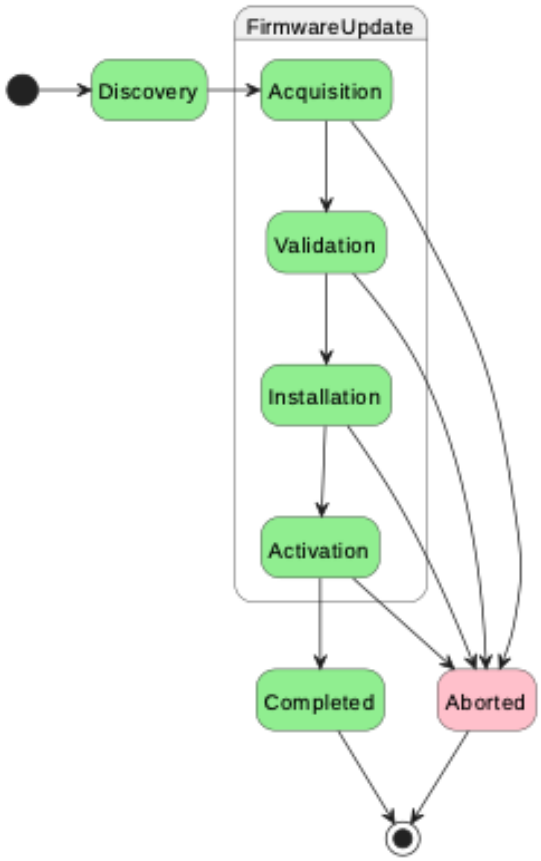
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4. Model

This specification extends the IPP model to support the Triggered Autonomous Firmware Update scenario described below.

The ability to update a device's Firmware is increasingly mandated by cybersecurity regulations. Figure 1 describes the phases involved in installing new Firmware.

Figure 1 - Firmware Update Phases

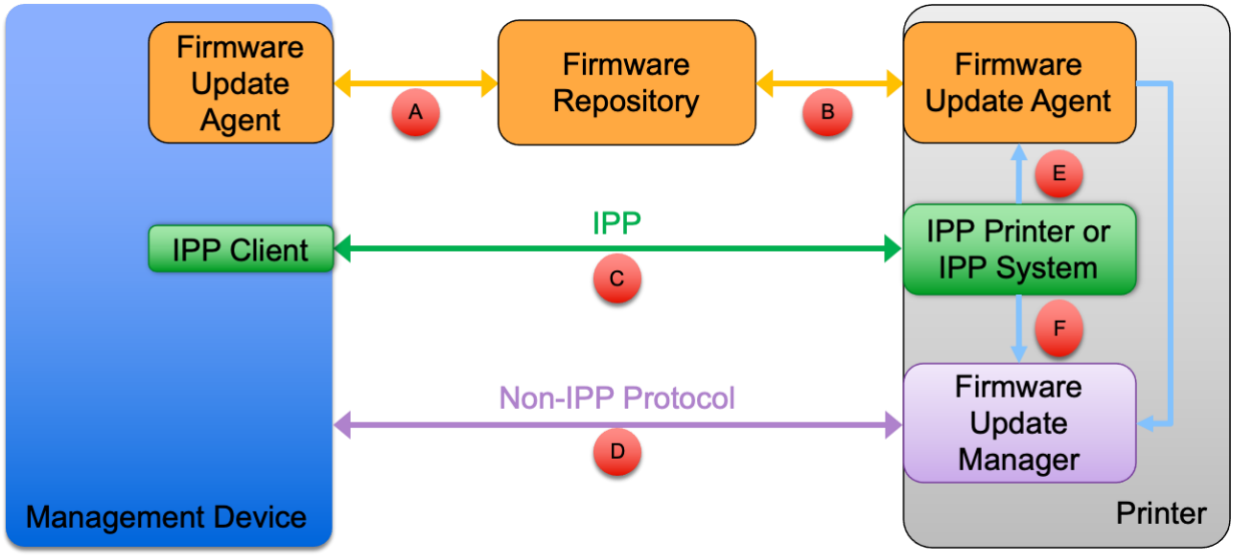


4.1 Firmware Update Interaction Types

Figure 2 shows the interactions between entities participating in a high-level conceptual model for various means of installing new Firmware in a Printer.

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Figure 2 - Firmware Update Conceptual Model

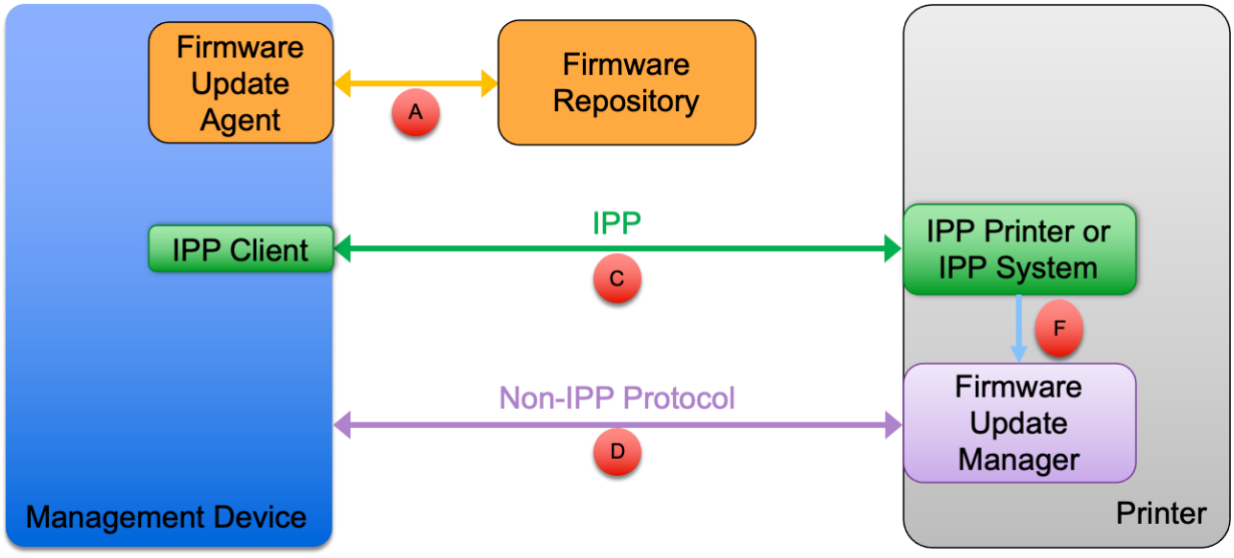


Several different interactions can use this model and are described below.

4.1.1 Directed Firmware Update

In this interaction, illustrated in Figure 3, a Firmware Update Agent running on a Management Device performs the Discovery phase by checking a Firmware Repository (A) for available new Firmware.

Figure 3 - Directed Firmware Update Model



The Management Device executes the Acquisition phase by downloading the new Firmware from the Firmware Repository (A), possibly performs the Validation phase, and then completes the Acquisition phase by submitting it to the Printer. If the Printer

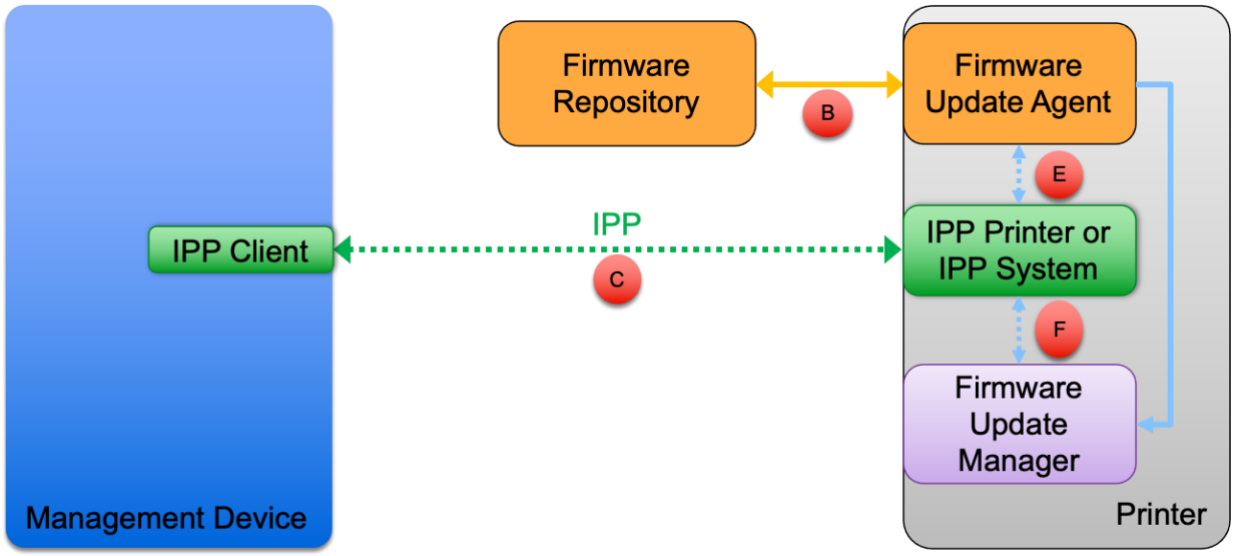
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implements the IPP System Service [PWG 5100.22] then the Management Device can submit the new Firmware via IPP (C). The Printer and Management Device might additionally implement some other protocol and system for completing the Acquisition phase (D). The Printer might perform its own Validation, and then performs the Installation, Activation and Cleanup phases.

4.1.2 Autonomous Firmware Update

In this interaction, illustrated in Figure 4, a Firmware Update Agent running in the Printer itself executes all phases.

Figure 4 - Autonomous Firmware Update Model



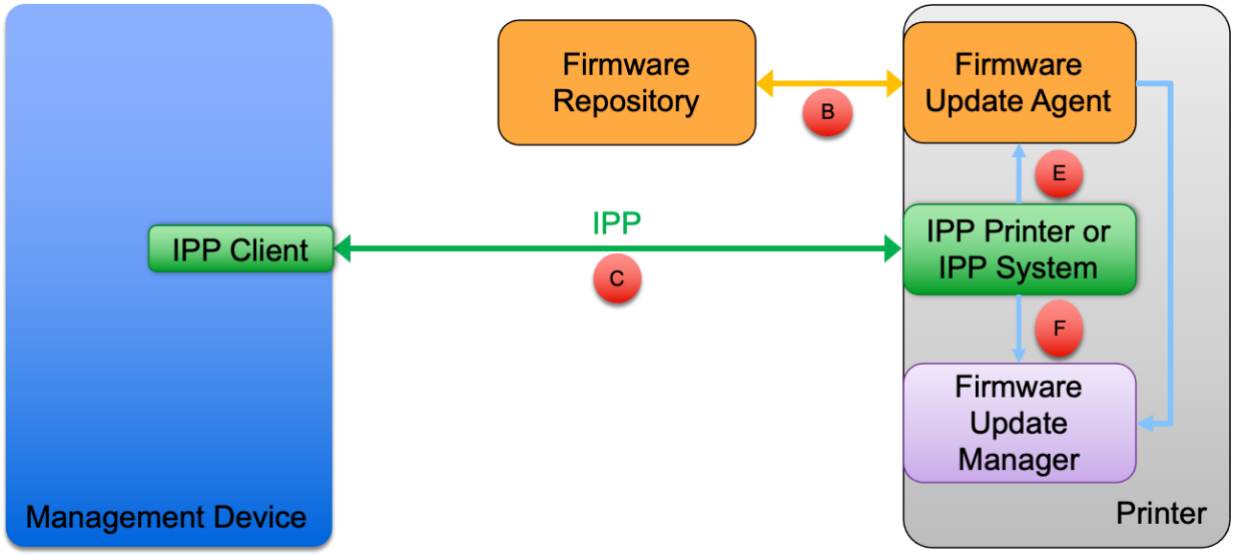
The Printer's Firmware Update Agent performs the Discovery phase by checking for available new Firmware from its Firmware Repository (B). If new Firmware is available, the Firmware Update Manager performs the remaining phases of a Firmware Update. The Printer can indicate its progress and status installing new Firmware via IPP (C). All these phases are executed within the Printer itself, without any user interaction, influence or control.

4.1.3 Triggered Autonomous Firmware Update

In this interaction, illustrated in Figure 5, IPP operations and attributes enhance and extend the facilities provided by the Autonomous Firmware Update model (section 4.1.2) to allow a Client to prompt the Printer to perform Discovery and Firmware Update on demand.

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Figure 5 - Triggered Autonomous Firmware Update Model

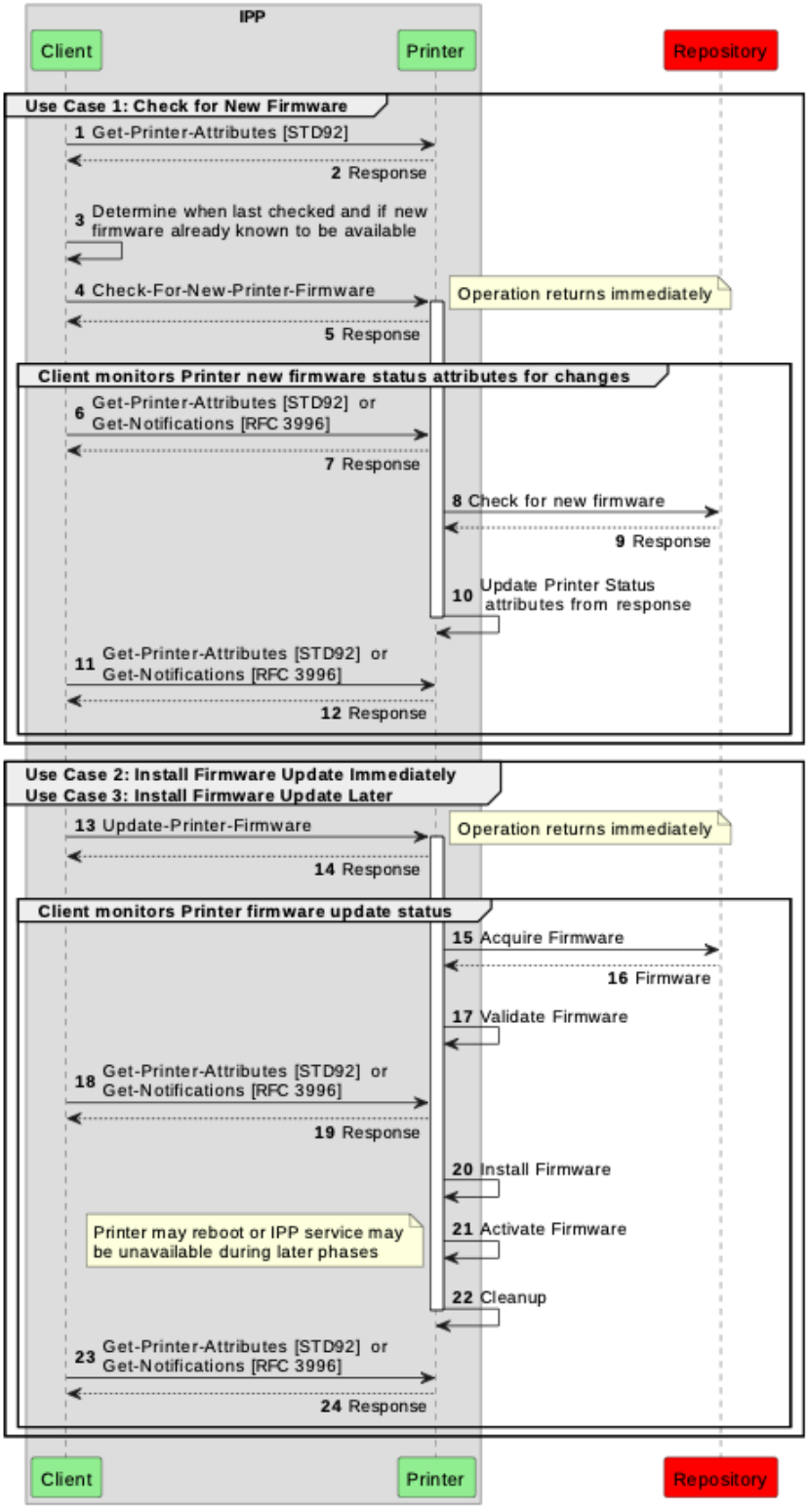


A Client uses the new "Check-For-New-Printer-Firmware" IPP operation (section 5.1) to request the Printer (C) to cause the Printer's Firmware Update Agent (E) to execute the Discovery phase (B). If the Printer's Firmware Update Agent discovers new Firmware is available, the Printer reports information about the available new Firmware via the attributes listed in Table 1.

A Client uses the new "Update-Printer-Firmware" IPP operation (section 5.2) to request the Printer to cause the Printer's Firmware Update Agent to perform the remaining phases of a Firmware Update. Figure 6 provides a UML sequence diagram illustrating the Client and Printer interactions and a successful Firmware Update.

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434 **Figure 6 - Triggered Autonomous Firmware Update Sequence Diagram**



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4.2 Current Firmware Status

A Printer implements the “printer-firmware-info-uri” (section 6.3.1) attribute, and the "printer-firmware-name", "printer-firmware-patches", "printer-firmware-string-version", and "printer-firmware-version" attributes [PWG 5100.13], to report information about its current Firmware.

4.3 New Firmware Availability

A Printer’s “printer-new-firmware-check-date-time” attribute (section 6.3.1) reports when the Printer last attempted to discover whether new Firmware is available from its Firmware Repository.

If the attempt succeeded and new Firmware is available, then the Printer adds the ‘new-firmware-available’ keyword (section 7.3.5) to its “printer-state-reasons” attribute (section 7.1) and describes the available new Firmware using the attributes listed in Table 1.

Table 1 - New Firmware Status Attributes

New Firmware Status Attribute	Section	Current Firmware Status Attribute
"printer-new-firmware-info-uri"	6.3.6	“printer-firmware-info-uri”
“printer-new-firmware-k-octets”	6.3.7	n/a
"printer-new-firmware-name"	6.3.8	“printer-firmware-name”
“printer-new-firmware-offline-time”	6.3.9	n/a
"printer-new-firmware-patches"	6.3.10	"printer-firmware-patches"
"printer-new-firmware-string-version"	6.3.11	"printer-firmware-string-version"
"printer-new-firmware-urgency"	6.3.12	n/a
"printer-new-firmware-version"	6.3.13	"printer-firmware-version"

If the attempt succeeded but no new Firmware is available, the Printer removes the 'new-firmware-available' keyword from its "printer-state-reasons" attribute and sets the values for the attributes listed in Table 1 to the ‘no-value’ out-of-band value.

If the attempt failed because the Printer was unable to connect to its Firmware Repository, then the Printer adds the ‘firmware-repository-unreachable’ keyword (section 7.3.2) to its “printer-state-reasons” attribute (section 7.1) and sets the values for the attributes listed in Table 1 to the ‘unknown’ out-of-band value.

If the attempt failed because the Printer was able to initiate a connection to its Firmware Repository but the connection was rejected by the Firmware Repository, then the Printer adds the ‘firmware-repository-access-error’ keyword (section 7.3.1) to its “printer-state-reasons” attribute and sets the values for the attributes listed in Table 1 to the ‘unknown’ out-of-band value.

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4.4 Requesting Printer to Check for New Firmware

A Client sends the Check-For-New-Firmware-Update operation (section 5.1) to request the Printer to check for new Firmware. The Printer responds immediately before completing the check. A Client can use the Get-Printer-Attributes [STD92] or Get-Notifications [RFC 3996] operations and the attributes listed in section 4.2 to discover the availability of new Firmware.

To protect itself and its Firmware Repository from Denial of Service (DoS) attacks, the Printer can provide cached values for the attributes listed in Table 1 for a reasonable amount of time. Additional guidance is provided in section 10.

4.5 Requesting Printer to Update Firmware

A Client sends the Update-Printer-Firmware operation (section 5.2) to request the Printer to install new Firmware if available. The request can include the “delay-update-until” attribute (section 6.1.1) or the “delay-update-until-date-time” attribute (section 6.1.2) to schedule when the Printer installs the available new Firmware. The Printer may download available new Firmware at any time.

A Client can monitor the Printer’s progress and status using the Get-Printer-Attributes [STD92] or Get-Notifications [RFC 3996] operations and examining the Printer’s “printer-state-reasons” attribute for keywords listed in section 7.3.

Note: The Printer can reboot during the process of installing new Firmware, causing the Printer to be temporarily unreachable.

4.6 Firmware Update Status

When a Printer begins a Firmware Update, it adds the ‘firmware-update-processing' keyword (section 7.3.14) to its “printer-state-reasons” Printer Status attribute (section 7.1) and begins performing the Firmware Update.

The Printer can indicate the phase in progress by adding the corresponding phase’s ‘firmware-update-<PHASE>’ keyword to its “printer-state-reasons” attribute. If that phase failed, then the Printer can indicate that failure by adding the corresponding phase’s ‘firmware-update-<PHASE>-failed’ keyword. Table 2 lists the status keywords for each phase.

Table 2 - Firmware Update “printer-state-reasons” Status Keywords

Phase	In Progress Keyword	Failure Keyword
Acquisition	firmware-update-acquisition	firmware-update-acquisition-failed
Validation	firmware-update-validation	firmware-update-validation-failed
Installation	firmware-update-installation	firmware-update-installation-failed
Activation	firmware-update-activation	firmware-update-activation-failed

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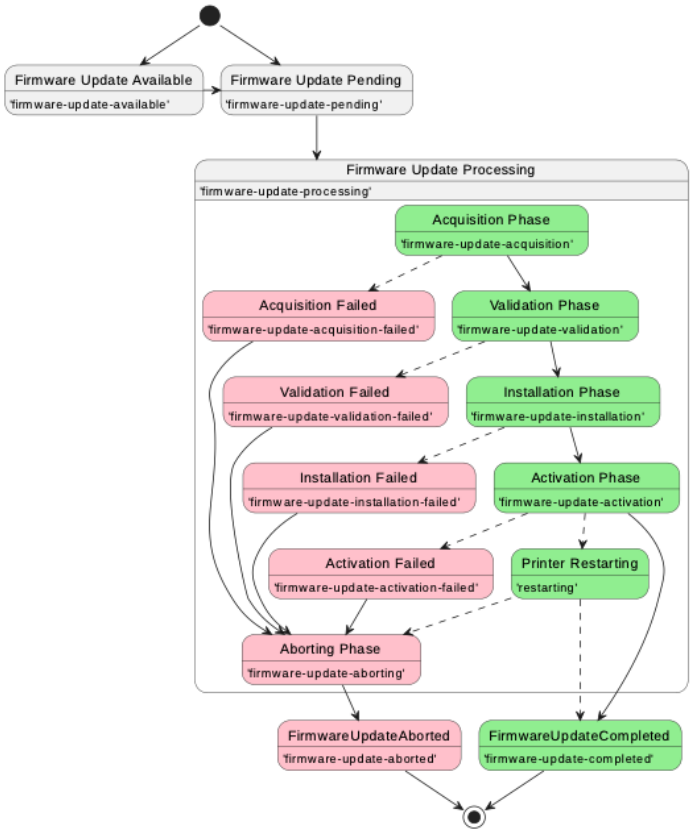
If the Firmware Update concludes unsuccessfully, then the Printer replaces the ‘firmware-update-processing’ keyword with the 'firmware-update-aborted' keyword (section 7.3.3), for a duration that is implementation specific.

If the Firmware Update concludes successfully, then the Printer replaces the 'firmware-update-processing' keyword listed by its “printer-state-reasons” attribute with the 'firmware-update-completed' keyword (section 7.3.10), for a duration that is implementation specific, and resets the values of the attributes listed in Table 1 to their default values and can stop listing them, as appropriate, until the Discovery phase is performed again and new Firmware is once again discovered.

If during the Firmware Update the Printer cannot accept new Jobs, then it sets its "printer-state" attribute [STD92] to 5 (‘stopped’) and sets the value of its “printer-is-accepting-jobs” Printer Status attribute [STD92] to ‘false’. If the Printer will be unreachable during the Firmware Update, then it will indicate the duration via the “printer-new-firmware-offline-time” attribute (section 6.3.9). When the Printer is ready to once again accept Jobs, it sets the value of its “printer-is-accepting-jobs” attribute to ‘true’ and sets the value of its “printer-state” attribute to 3 (‘idle’).

Figure 7 illustrates the transitions between states and phases, and their corresponding “printer-state-reasons” keywords.

Figure 7 - Firmware Update Phases and 'printer-state-reasons' keywords



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4.7 Selecting a Printer Firmware Repository

The “printer-firmware-repository-uri-configured” attribute (section 6.2.4) reports the URI of the Printer’s currently configured Firmware Repository. The “printer-firmware-repository-uri-supported” attribute (section 6.2.5) lists the set of Firmware Repository URIs from which the “printer-firmware-repository-uri-configured” can be set.

The Printer’s "printer-settable-attributes-supported" attribute [RFC 3380] will list ‘printer-firmware-repository-uri-configured’ and/or ‘printer-firmware-repository-uri-supported’ if either of those attributes can be configured using IPP via a Set-Printer-Attributes operation [RFC 3380]. An authorized Client can select a different Firmware Repository using the Set-Printer-Attributes operation to set the value of "printer-firmware-repository-uri-configured" attribute to one of the values listed by “printer-firmware-repository-uri-supported”. An authorized Client can configure which Firmware Repositories are allowed by setting the “printer-new-firmware-update-repository-supported” attribute using the Set-Printer-Attributes operation.

An Authorized Client can use the Get-Printer-Supported-Values operation [RFC 3380] to obtain the factory default list of values for the “printer-firmware-repository-uri-supported” attribute.

4.8 Updating the Printer Firmware Update Feature Policy

The "printer-new-firmware-policy-configured" attribute (section 6.2.2) attribute reports the Printer’s currently configured policy for checking for new Firmware. The “printer-new-firmware-policy-supported” attribute (section 6.2.3) lists the set of policies supported by the Printer.

The Printer’s "printer-settable-attributes-supported" attribute [RFC 3380] will list ‘printer-new-firmware-policy-configured’ if the Printer supports configuring the policy using IPP via a Set-Printer-Attributes operation [RFC 3380]. An authorized Client can configure the Printer's Firmware Update policy by setting the value of the “printer-new-firmware-policy-configured” attribute to one of the values listed by “printer-new-firmware-policy-supported”.

5. New Operations

5.1 Check-For-New-Printer-Firmware

This RECOMMENDED operation causes the Printer to perform the Discovery phase by checking its Firmware Repository for available new Firmware. The Printer responds immediately before performing the check. If the Printer discovers that new Firmware is available, then it describes the available new Firmware via the attributes described in section 4.2.

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The Authenticated User performing this operation MUST be either an Operator or Administrator for the Printer. Otherwise, the Printer MUST reject the operation and return the 'client-error-forbidden', 'client-error-not-authenticated', or 'client-error-not-authorized' status code, as appropriate.

5.1.1 Check-For-New-Printer-Firmware Request

The following groups of attributes are supplied in a Check-For-New-Printer-Firmware request:

Group 1: Operation Attributes

"attributes-charset" (charset) and "attributes-natural-language" (naturalLanguage):

The Client MUST supply and the Printer MUST support both attributes, as described in [STD92] Section 4.1.4.1.

"printer-uri" (uri):

The Client MUST supply and the Printer MUST support this attribute, which specifies the target for this operation, as described in [STD92] Section 4.1.5.

"requesting-user-name" (name(MAX)):

The Client SHOULD supply and the Printer MUST support this attribute, as described in [STD92] Section 9.3.

“requesting-user-uri” (uri):

The Client SHOULD supply and the Printer MUST support this attribute, as described in [PWG 5100.13] section 5.1.6.

5.1.2 Check-For-New-Printer-Firmware Response

The following groups of attributes are supplied in a Check-For-New-Printer-Firmware response:

Group 1: Operation Attributes

"attributes-charset" (charset) and "attributes-natural-language" (naturalLanguage):

The Client MUST support and the Printer MUST supply both attributes, as described in [STD92] Section 4.1.4.1.

Status Message:

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In addition to the REQUIRED status-code returned, the response MAY include the "status-message" (text(255)) and/or the "detailed-status-message" (text(MAX)) operation attribute as described in [STD92] Appendix B and Section 4.1.6.

Group 2: Printer Attributes

"printer-new-firmware-last-checked" (dateTime)

The Printer MAY supply this attribute, as defined in section 6.3.1.

"printer-new-firmware-k-octets" (integer(1:MAX) | unknown)

The Printer MAY supply this attribute, as defined in section 6.3.7.

"printer-new-firmware-support-uri" (uri)

The Printer MAY supply this attribute, as defined in section 6.3.6.

"printer-new-firmware-urgency"

The Printer MAY supply this attribute, as defined in section 6.3.12

"printer-new-firmware-version"

The Printer MAY supply this attribute, as defined in section 6.3.13

"printer-new-firmware-string-version"

The Printer MAY supply this attribute, as defined in section 6.3.11

5.2 Update-Printer-Firmware

This REQUIRED operation causes the Printer to attempt the Discovery phase and, if new Firmware is available, to perform the remainder of the Firmware Update phases. The Printer responds before beginning the Firmware Update.

The Client sends the Update-Printer-Firmware request over a secure connection [STD92] as per section 10. If the Printer receives an Update-Printer-Firmware request over an insecure connection (e.g. HTTP without TLS), then the Printer MUST respond with an HTTP 426 response to upgrade the connection to TLS [RFC 2817] [RFC9110] or reject the request.

The Authenticated User performing this operation MUST be an Operator or Administrator for the Printer. Otherwise, the Printer MUST reject the operation and return the 'client-error-forbidden', 'client-error-not-authenticated', or 'client-error-not-authorized' status code, as appropriate.

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5.2.1 Scheduled Updates

If the operation is successful and the Client supplies either the “delay-update-until” or “delay-update-until-time” operation attributes, then the Printer does not begin performing the Firmware update until the specified time and includes the current “firmware-update-delayed-until” (section 6.3.1) and “firmware-update-delayed-until-time” (section 6.3.5) attributes in the response. If new Firmware installation was already scheduled with a delay, and changes to that are desired, then a Client can perform a new Update-Printer-Firmware request to supply a new value.

A Client MUST NOT supply both the "delay-update-until" and "delay-update-until-time" (section 6.1.2) operation attributes in an Update-Printer-Firmware request. If a Client supplies both attributes, then the Printer MUST reject the request and return the 'client-error-conflicting-attributes' status code. While the specified time period has not yet arrived, the Printer includes the 'firmware-update-delay-until-specified' value in the Printer's "printer-state-reasons" Printer Status attribute (section 7.1).

5.2.2 Update-Printer-Firmware Request

The following groups of attributes are supplied in an Update-Printer-Firmware request:

Group 1: Operation Attributes

"attributes-charset" (charset) and "attributes-natural-language" (naturalLanguage):

The Client MUST supply and the Printer MUST support both of these attributes, as described in [STD92] Section 4.1.4.1.

"printer-uri" (uri):

The Client MUST supply and the Printer MUST support this attribute, which specifies the target for this operation, as described in [STD92] Section 4.1.5.

"requesting-user-name" (name(MAX)):

The Client MAY supply and the Printer MUST support this attribute, as described in [STD92] Section 9.3.

"delay-update-until" (type2 keyword | name(MAX))

The Client MAY supply and the Printer MAY support this attribute, as defined in section 6.1.1.

"delay-update-until-time" (dateTime)

The Client MAY supply and the Printer MAY support this attribute, as defined in section 6.1.2.

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5.2.3 Update-Printer-Firmware Response

The following groups of attributes are supplied in an Update-Printer-Firmware response:

Group 1: Operation Attributes

"attributes-charset" (charset) and "attributes-natural-language" (naturalLanguage):

The Client MUST support and the Printer MUST supply both of these attributes, as described in [STD92] Section 4.1.4.1.

Status Message:

In addition to the REQUIRED status-code returned, the response MAY include the "status-message" (text(255)) and/or the "detailed-status-message" (text(MAX)) operation attribute as described in [STD92] Appendix B and Section 4.1.6.

Group 2: Printer Attributes

“printer-new-firmware-delayed-until” (type2 keyword)

The Printer MAY supply this attribute, as described in section 6.3.1.

“printer-new-firmware-delayed-until-time” (dateTime | no-value)

The Printer MAY supply this attribute, as described in section 6.3.5.

Group 3: Unsupported Attributes

See [STD92] Section 4.1.7 for details on returning unsupported attributes.

6. New Attributes

6.1 Operation Attributes

6.1.1 delay-update-until (type2 keyword)

This attribute specifies a named time period when the Printer will install new Firmware as requested in an Update-Printer-Firmware operation (section 5.2). Table 3 lists the keyword values defined in this specification for use with this attribute.

Table 3 – Keywords for "delay-update-until"

Keyword	Description
‘no-delay’	Install new Firmware immediately.
'indefinite'	Delay new Firmware install indefinitely.
'day-time'	Delay new Firmware install until daylight.

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Keyword	Description
'evening'	Delay new Firmware install until the evening.
'night'	Delay new Firmware install until the night.
'second-shift'	Delay new Firmware install until the second work shift.
'third-shift'	Delay new Firmware install until the third work shift.
'weekend'	Delay new Firmware install until the weekend.

658 ### 6.1.2 delay-update-until-date-time (dateTime)

659 This attribute specifies a precise date and time when the Printer will install the new
660 Firmware as requested in an Update-Printer-Firmware operation (section 5.2).

661 ## 6.2 Printer Description Attributes

662 ### 6.2.1 delay-update-until-supported (1setOf type2 keyword)

663 This REQUIRED attribute indicates the values the Printer supports for the “delay-update-
664 until” attribute (section 6.1.1).

665 ### 6.2.2 printer-new-firmware-policy-configured (type2 keyword)

666 This REQUIRED attribute indicates the currently configured policy for the Triggered
667 Autonomous Firmware Update feature (section 4.1.3). Table 4 lists the keywords defined
668 in this specification.

669 If the Printer supports configuring this attribute via IPP, then it will list this attribute in its
670 “printer-settable-attributes-supported” attribute [RFC 3380].

671 **Table 4 – Keywords for "printer-new-firmware-policy-configured"**

Keyword	Description
‘auto’	Printer automatically performs the Discovery, Acquisition, Validation, Installation, Activation, and Cleanup phases.
‘check’	Printer automatically performs the Discovery phase, but refrains performing the Acquisition, Validation, Installation, Activation, or Cleanup phases until it has received an Update-Printer-Firmware operation (section 5.2).
‘directed’	Printer performs the Discovery phase only in response to a Check-For-New-Printer-Firmware operation (section 5.1). Printer performs the Acquisition, Validation, Installation, Activation, and Cleanup phases only in response to the operations defined in section 5.
‘download’	Printer automatically performs the Discovery, Acquisition and Validation phases, but refrains from performing the Installation, Activation, or Cleanup phases until it has received an Update-Printer-Firmware operation (section 5.2).

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672 **6.2.3 printer-new-firmware-policy-supported (1setOf type2 keyword)**

673 This REQUIRED attribute lists the supported policies for Triggered Autonomous Firmware
674 Update (section 4). If the Printer supports configuring this attribute via IPP, then it will list
675 this attribute in its “printer-settable-attributes-supported” attribute [RFC 3380].

676 **6.2.4 printer-firmware-repository-uri-configured (uri)**

677 This CONDITIONALLY REQUIRED attribute provides the currently configured URI of the
678 Firmware Repository the Printer uses to check for new Firmware, either on its own or in
679 response to a Check-For-New-Printer-Firmware operation (section 5.1).

680 If the Printer supports changing this attribute’s value, then the Printer MUST list this
681 attribute's name in its "printer-settable-attributes-supported" attribute [RFC 3380], MUST
682 support the Set-Printer-Attributes operation [RFC 3380], MUST support the “printer-
683 firmware-repository-uri-supported” attribute (section 6.2.5), and MUST only accept a value
684 that listed by the Printer’s “printer-firmware-repository-uri-supported” attribute.

685 **6.2.5 printer-firmware-repository-uri-supported (1setOf uri)**

686 This attribute lists the Firmware Repository URIs that can be used to configure the
687 Printer’s “printer-firmware-repository-uri-configured” attribute (section 6.2.2).

688 If the Printer supports changing this attribute’s values, then the Printer MUST support the
689 Get-Printer-Supported-Values operation [RFC 3380] and the value returned for this
690 attribute MUST include the manufacturer default Firmware Repository URI.

691 ## 6.3 Printer Status Attributes

692 **6.3.1 printer-firmware-info-uri (uri)**

693 This REQUIRED attribute provides a 'http:' or 'https:' scheme URI to user readable content
694 describing the currently installed Firmware.

695 **6.3.2 printer-firmware-update-date-time (dateTime | no-value)**

696 This REQUIRED attribute reports the time when the currently running Firmware was
697 installed. A value of ‘no-value’ indicates the Firmware was installed at the time the Printer
698 was manufactured.

699 **6.3.3 printer-new-firmware-check-date-time (dateTime | no-value | unknown)**

700 This REQUIRED attribute reports the value reported by the “printer-current-time” attribute
701 [STD92] when the Printer last attempted to check for new Firmware, either on its own or in
702 response to a Check-For-New-Printer-Firmware operation (section 5.1). The value is
703 updated even when the Printer is unable to contact the configured Firmware Repository or

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obtain actionable information. If the Printer has never checked for new Firmware, then the Printer MUST return the ‘no-value’ out-of-band value.

Note: If the Printer’s clock is not set when the new Firmware check is run, then the Printer’s “printer-current-time” attribute [STD92] will report the ‘unknown’ out-of-band value, and in such circumstances this attribute will also report ‘unknown’.

6.3.4 printer-new-firmware-delayed-until (type2 keyword)

This attribute indicates the delay set from the "delay-update-until" operation attribute (section 6.1.1) received in a previous Update-Printer-Firmware operation request (section 5.2) that was successful and accepted by the Printer. Table 3 lists the keyword values defined in this specification this attribute can hold. A Printer will omit this attribute if the most recent Update-Printer-Firmware request was successful and did not include “delay-update-until”.

6.3.5 printer-new-firmware-delayed-until-date-time (dateTime)

This attribute indicates the delay set from the "delay-update-until-date-time" operation attribute (section 6.1.2) received in a previous Update-Printer-Firmware operation request (section 5.2) that was successful and accepted by the Printer. A Printer will omit this attribute if the most recent Update-Printer-Firmware request was successful and did not include “delay-update-until-date-time”.

6.3.6 printer-new-firmware-info-uri (uri | no-value | unknown)

This REQUIRED attribute provides a 'http:' or 'https:' URI to user readable content describing the available new Firmware. If the Printer has never checked for new Firmware, then the Printer returns the ‘no-value’ out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the ‘unknown’ out-of-band value.

6.3.7 printer-new-firmware-k-octets (1setOf integer(1:MAX) | no-value | unknown)

This REQUIRED attribute indicates the size of the new Firmware, measured in kiB (1024 octets). If the size is unknown, then the Printer returns the 'unknown' out-of-band value. If the Printer has never checked for new Firmware, then the Printer returns the ‘no-value’ out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the ‘unknown’ out-of-band value.

6.3.8 printer-new-firmware-name (1setOf name(MAX) | no-value | unknown)

This REQUIRED attribute provides the names of the new Firmware, in a format matching the Printer's “printer-firmware-name” attribute [PWG 5100.13]. If the Printer has never checked for new Firmware, then the Printer returns the ‘no-value’ out-of-band value. If the

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Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the ‘unknown’ out-of-band value.

6.3.9 printer-new-firmware-offline-time (integer(0:MAX))

This attribute indicates the number of seconds the Printer will typically be unreachable during a successful Firmware Update. If the Printer does not support this attribute, then the Client can assume that the Printer will respond to IPP requests during the entire Firmware Update.

6.3.10 printer-new-firmware-patches (1setOf text(MAX) | no-value | unknown)

This REQUIRED attribute provides the set of patches to be applied to each of the Printer firmware components named by the "printer-new-firmware-name" Printer Status attribute, in a format matching the Printer's “printer-firmware-patches” attribute [PWG 5100.13]. If the Printer has never checked for new Firmware, then the Printer returns the ‘no-value’ out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the ‘unknown’ out-of-band value.

6.3.11 printer-new-firmware-string-version (1setOf text(MAX) | no-value | unknown)

This REQUIRED attribute provides the string versions of the values provided by the "printer-new-firmware-version" attribute (section 6.3.13) , in a format matching the Printer's “printer-firmware-string-version” attribute [PWG 5100.13]. If the Printer has never checked for new Firmware, then the Printer returns the ‘no-value’ out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the ‘unknown’ out-of-band value.

6.3.12 printer-new-firmware-urgency (type2 keyword | no-value | unknown)

This REQUIRED attribute indicates the urgency of the new Firmware. Table 5 lists the keywords defined for this attribute. If the Printer has never checked for new Firmware, then the Printer returns the ‘no-value’ out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the ‘unknown’ out-of-band value.

Table 5 – Keywords for "printer-new-firmware-urgency"

Keyword	Description
‘bugfix’	The available new Firmware provides minor bug fixes
‘feature’	The available new Firmware provides new features
‘security’	The available new Firmware provides security updates

6.3.13 printer-new-firmware-version (1setOf octetString(MAX) | no-value | unknown)

This REQUIRED attribute reports the latest firmware version available for the Printer by the Printer's manufacturer, in a format matching the Printer's "printer-firmware-version"

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attribute [PWG 5100.13]. If the Printer has never checked for new Firmware, then the Printer returns the ‘no-value’ out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the ‘unknown’ out-of-band value.

7. New Values for Existing Attributes

7.1 ipp-features-supported

This specification defines the new ‘fwupdate’ keyword for the "ipp-features-supported" attribute [PWG 5100.13] to indicate support for the Triggered Autonomous Firmware Update feature defined in this specification.

7.2 operations-supported (1setOf type2 enum)

Table 6 lists the following new keywords for the "operations-supported" attribute [STD92] defined by this specification.

Table 6 – Enum values for "operations-supported"

Enum	Description
0x006B	Check-For-New-Printer-Firmware
0x006C	Update-Printer-Firmware

7.3 printer-state-reasons (1setOf type2 keyword)

This section defines new keywords for the "printer-state-reasons" attribute [STD92]. These are used to indicate what phase of the Firmware installation process is occurring currently, as described in Figure 1. Keywords are defined for each phase to indicate if the phase is “in progress”, ended in “failure” or ended in “success”. Other keywords indicate specific failure conditions that benefit from more detail or other status conditions possibly occurring at other times.

7.3.1 firmware-repository-access-error

This REQUIRED keyword indicates that the Firmware Repository is reachable but that the new Firmware could not be downloaded. Possible reasons could include TLS server certificate rejection, cipher suite rejection, missing file, or unauthorized / forbidden access.

When this keyword is added, the ‘firmware-repository-unreachable’ keyword (section 7.3.2) and the ‘new-firmware-available’ keyword (section 7.3.5) MUST be removed, if present.

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7.3.2 firmware-repository-unreachable

This REQUIRED keyword indicates that the Printer was unable to connect to the currently configured Firmware Repository as specified by the Printer's “printer-firmware-repository-uri-configured” attribute (section 6.2.4).

When this keyword is added, the ‘firmware-repository-access-error’ keyword (section 7.3.1) and the ‘new-firmware-available’ keyword (section 7.3.5) MUST be removed, if present.

7.3.3 firmware-update-aborted

This REQUIRED keyword indicates the Printer has recently attempted to update its Firmware but that attempt failed.

7.3.4 firmware-update-aborting

This REQUIRED keyword indicates the Printer has recently attempted to update its Firmware but that attempt failed and the Printer is in the process of reverting to its previous state.

7.3.5 firmware-update-acquisition

This RECOMMENDED keyword indicates that the Printer is currently performing the Acquisition phase of a Firmware Update.

7.3.6 firmware-update-acquisition-failed

This RECOMMENDED keyword indicates that the Printer’s most recent attempt to perform the Acquisition phase of a Firmware Update has failed.

7.3.7 firmware-update-activation

This RECOMMENDED keyword indicates that the Printer is currently performing the Activation phase of a Firmware Update.

7.3.8 firmware-update-activation-failed

This RECOMMENDED keyword indicates that the Printer’s most recent attempt to perform the Activation phase of a Firmware Update has failed.

7.3.9 firmware-update-available

This REQUIRED keyword indicates the Printer has discovered that new Firmware is available from its Firmware Repository.

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When this keyword is added, the ‘firmware-repository-access-error’ keyword (section 7.3.1) and the ‘firmware-repository-unreachable’ keyword (section 7.3.2) MUST be removed, if present.

7.3.10 firmware-update-completed

This REQUIRED keyword indicates the Printer has recently succeeded at updating its Firmware.

7.3.11 firmware-update-installation

This RECOMMENDED keyword indicates that the Printer is currently performing the Install phase of a Firmware Update.

7.3.12 firmware-update-installation-failed

This RECOMMENDED keyword indicates that the Printer’s most recent attempt to perform the Installation phase of a Firmware Update has failed.

7.3.13 firmware-update-pending

This REQUIRED keyword indicates that the Printer will soon begin a Firmware Update.

7.3.14 firmware-update-processing

This REQUIRED keyword indicates the Printer is currently in the process of updating its Firmware.

7.3.15 firmware-update-validation

This RECOMMENDED keyword indicates that the Printer is currently performing the Validation phase of a Firmware Update.

7.3.16 firmware-update-validation-failed

This RECOMMENDED keyword indicates that the Printer’s most recent attempt to perform the Validation phase of a Firmware Update has failed.

7.3.17 restarting

This REQUIRED keyword indicates that the Printer will soon restart, and during that time will likely be unreachable.

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8. Conformance Requirements

8.1 Printer Conformance Requirements

Printers conforming to this Specification MUST support:

1. The model described in section 4.1.3;
2. The required operations defined in section 5;
3. The "printer-firmware-name", "printer-firmware-patches", "printer-firmware-string-version", and "printer-firmware-version" attributes [PWG 5100.13];
4. The required attributes and values defined in section 6;
5. The required additional values defined in section 7;
6. The internationalization considerations defined in section 9; and
7. The security and privacy considerations defined in section 10.

8.2 Client Conformance Requirements

Clients conforming to this Specification MUST support:

1. The model described in section 4.1.3;
2. The required operations defined in section 5;
3. The "printer-firmware-name", "printer-firmware-patches", "printer-firmware-string-version", and "printer-firmware-version" attributes [PWG 5100.13];
4. The required attributes and values defined in section 6;
5. The required additional values defined in section 7;
6. The internationalization considerations defined in section 9; and
7. The security and privacy considerations defined in section 10.

9. Internationalization Considerations

For interoperability and basic support for multiple languages, conforming implementations MUST support:

1. The Universal Character Set (UCS) Transformation Format -- 8 bit (UTF-8) [STD63] encoding of Unicode [UNICODE] [ISO10646]; and
2. The Unicode Format for Network Interchange [RFC 5198] which requires transmission of well-formed UTF-8 strings and recommends transmission of normalized UTF-8 strings in Normalization Form C (NFC) [UAX15].

Unicode NFC is defined as the result of performing Canonical Decomposition (into base characters and combining marks) followed by Canonical Composition (into canonical composed characters wherever Unicode has assigned them).

WARNING – Performing normalization on UTF-8 strings received from Clients and subsequently storing the results (e.g., in Job objects) could cause false negatives in Client

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searches and failed access (e.g., to Printers with percent-encoded UTF-8 URIs now 'hidden').

Implementations of this specification SHOULD conform to the following standards on processing of human-readable Unicode text strings, see:

- Unicode Bidirectional Algorithm [UAX9] – left-to-right, right-to-left, and vertical
- Unicode Line Breaking Algorithm [UAX14] – character classes and wrapping
- Unicode Normalization Forms [UAX15] – especially NFC for [RFC 5198]
- Unicode Text Segmentation [UAX29] – grapheme clusters, words, sentences
- Unicode Identifier and Pattern Syntax [UAX31] – identifier use and normalization
- Unicode Collation Algorithm [UTS10] – sorting
- Unicode Locale Data Markup Language [UTS35] – locale databases

Implementations of this specification are advised to also review the following informational documents on processing of human-readable Unicode text strings:

- Unicode Character Encoding Model [UTR17] – multi-layer character model
- Unicode Character Property Model [UTR23] – character properties
- Unicode Conformance Model [UTR33] – Unicode conformance basis

10. Security and Privacy Considerations

Provide security and privacy considerations for this specification, such as the following.

The IPP extensions defined in this document require the same security and privacy considerations as defined in the Internet Printing Protocol/1.1 [STD92].

Implementations of this specification SHOULD conform to the following standard on processing of human-readable Unicode text strings, see:

- Unicode Security Mechanisms [UTS39] – detecting and avoiding security attacks
- Implementations of this specification are advised to also review the following informational document on processing of human-readable Unicode text strings:
- Unicode Security FAQ [UNISECFAQ] – common Unicode security issues

Implementations of this specification SHOULD comply with the following:

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- NIST SP800-193 Platform Firmware Resiliency Guidelines (May 2018) https://csrc.nist.gov/pubs/sp/800/193/final

- ITU-T X.1373 Secure Software Update Capability for ITS (March 2024) https://www.itu.int/ITU-T/recommendations/rec.aspx?rec=15664

Printers implementing this specification SHOULD comply with the following guidance:

1. Communications between the Printer and the Firmware Repository is protected in transit (i.e., TLS, DTLS, etc.);

2. Firmware files are cryptographically signed, and Printers validate these cryptographic signatures before proceeding with installation;

3. Identifiers used to track information about existing Firmware versions, authorize access to new Firmware, and/or monitor download and installation performance do not include Personally Identifiable Information (PII).

11. Implementation and Deployment Considerations

This specification defines only an IPP interface to allow a Client to interact with a subset of a Printer's own Firmware update capability. Though this helps to facilitate a Firmware update user experience using IPP, other elements of the user experience are the sole responsibilities of the Client, the Printer and the Firmware Repository.

A well-implemented Client supporting this specification SHOULD:

1. Present UI elements appropriately adjacent to other Printer basic information that lists the Printer's "printer-firmware-xxx" attributes defined in [PWG 5100.13] and this specification, including the "printer-firmware-info-uri" (section 6.3.1) so the user has ready access to the release notes and other information about the currently installed Firmware;

2. Update any cached Printer information with the latest settings, if caching is used, so that the Printer's current capabilities and features are accurately reflected in the Client's UI, and notification that the Firmware was updated;

3. Present UI elements that can appropriately present the Printer's status concerning the current Firmware as well as any unsuccessful attempts to install available new Firmware. Also, present UI elements to provide the user with easy access to the vendor's support page for that Printer model, which is provided by the Printer's "printer-more-info-manufacturer" attribute [STD92].

A well-implemented Printer supporting this specification SHOULD:

1. Provide UI elements (physical or graphical) to indicate failed or incomplete installation of new Firmware on the Printer's console;

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2. Support the “printer-more-info-manufacturer” Printer Description attribute, with the value being a URI that points to the support page for the Printer’s particular model; and

3. Host a locally cached copy of the release notes for the currently installed Firmware.

A well-implemented Firmware Repository supporting this specification SHOULD provide the following:

1. Provide a “Release Notes” document describing the currently available new Firmware at a publicly accessible URL.

12. IANA Considerations

12.1 Attribute Registrations

The attributes defined in this specification will be published by IANA according to the procedures in the Internet Printing Protocol/1.1 [STD92] in the following file:

https://www.iana.org/assignments/ipp-registrations

The registry entries will contain the following information:

```
Operation Attributes:                                          Reference
-----------------------------                                  ---------
delay-update-until (type2 keyword)                             [FWUPDATE]
delay-update-until-date-time (dateTime)                        [FWUPDATE]

Printer Description Attributes:                                Reference
-----------------------------                                  ---------
delay-update-until-supported (1setOf type2 keyword)            [FWUPDATE]
printer-new-firmware-policy-configured (type2 keyword)         [FWUPDATE]
printer-new-firmware-policy-supported (1setOf type2 keyword)
                                                               [FWUPDATE]
printer-firmware-repository-uri-configured (uri)               [FWUPDATE]
printer-firmware-repository-uri-supported (1setOf uri)         [FWUPDATE]


Printer Status Attributes:                                     Reference
-----------------------------                                  ---------
printer-firmware-info-uri (uri)                                [FWUPDATE]
printer-firmware-update-date-time (dateTime | no-value)        [FWUPDATE]
printer-new-firmware-check-date-time (dateTime | no-value | unknown)
                                                               [FWUPDATE]
printer-new-firmware-delayed-until (type2 keyword | no-value)
                                                               [FWUPDATE]
printer-new-firmware-delayed-until-date-time (dateTime | no-value)
                                                               [FWUPDATE]
printer-new-firmware-info-uri (uri | no-value | unknown)       [FWUPDATE]
printer-new-firmware-k-octets (1setOf integer(1:MAX) | no-value | unknown)
                                                               [FWUPDATE]
```

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```
printer-new-firmware-name (1setOf name(MAX) | no-value | unknown)
                                                                [FWUPDATE]
printer-new-firmware-patches (1setOf text(MAX) | no-value | unknown)
                                                                [FWUPDATE]
printer-new-firmware-string-version (1setOf text(MAX) | no-value |
unknown)                                                        [FWUPDATE]
printer-new-firmware-urgency (type2 keyword | no-value | unknown)
                                                                [FWUPDATE]
printer-new-firmware-version (1setOf octetString(MAX) | no-value |
unknown)                                                        [FWUPDATE]
```

12.2 Type2 keyword Registrations

The attributes defined in this specification will be published by IANA according to the procedures in the Internet Printing Protocol/1.1 [STD92] in the following file:

https://www.iana.org/assignments/ipp-registrations

The registry entries will contain the following information:

```
Attributes (attribute syntax)
  Keyword Attribute Value                                   Reference
  -----------------------                                   ---------
delay-update-until (type2 keyword)                          [FWUPDATE]
  no-hold                                                   [FWUPDATE]
  indefinite                                                [FWUPDATE]
  day-time                                                  [FWUPDATE]
  evening                                                   [FWUPDATE]
  night                                                     [FWUPDATE]
  second-shift                                              [FWUPDATE]
  third-shift                                               [FWUPDATE]
  weekend                                                   [FWUPDATE]
ipp-features-supported (1setOf type2 keyword)               [PWG5100.13]
  fwupdate                                                  [FWUPDATE]
printer-new-firmware-policy-configured (type2 keyword)      [FWUPDATE]
  auto                                                      [FWUPDATE]
  check                                                     [FWUPDATE]
  directed                                                  [FWUPDATE]
  download                                                  [FWUPDATE]
printer-new-firmware-delayed-until (type2 keyword | no-value)
                                                            [FWUPDATE]
  <Any "delay-update-until" keyword value>                  [FWUPDATE]
printer-new-firmware-urgency (type2 keyword | no-value | unknown)
                                                            [FWUPDATE]
  bugfix                                                    [FWUPDATE]
  feature                                                   [FWUPDATE]
  security                                                  [FWUPDATE]
printer-state-reasons (type2 keyword)                       [RFC8011]
  firmware-repository-access-error                          [FWUPDATE]
  firmware-repository-unreachable                           [FWUPDATE]
  firmware-update-aborted                                   [FWUPDATE]
  firmware-update-aborting                                  [FWUPDATE]
  firmware-update-acquisition                               [FWUPDATE]
```

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```
  firmware-update-acquisition-failed                    [FWUPDATE]
  firmware-update-activation                            [FWUPDATE]
  firmware-update-activation-failed                     [FWUPDATE]
  firmware-update-available                             [FWUPDATE]
  firmware-update-completed                             [FWUPDATE]
  firmware-update-installation                          [FWUPDATE]
  firmware-update-installation-failed                   [FWUPDATE]
  firmware-update-pending                               [FWUPDATE]
  firmware-update-processing                            [FWUPDATE]
  firmware-update-requested                             [FWUPDATE]
  firmware-update-validation                            [FWUPDATE]
  firmware-update-validation-failed                     [FWUPDATE]
  restarting                                            [FWUPDATE]
```

12.3 Type2 enum Registrations

The attributes defined in this specification will be published by IANA according to the procedures in the Internet Printing Protocol/1.1 [STD92] in the following file:

https://www.iana.org/assignments/ipp-registrations

The registry entries will contain the following information:

```
Attributes (attribute syntax)
  Enum Value  Enum Symbolic Name                        Reference
  ----------  ------------------                        ---------
operations-supported (1setOf type2 enum)                [RFC8011]
  0x006B      Check-For-New-Printer-Firmware            [FWUPDATE]
  0x006C      Update-Printer-Firmware                   [FWUPDATE]
```

12.4 Operation Registrations

The attributes defined in this specification will be published by IANA according to the procedures in the Internet Printing Protocol/1.1 [STD92] in the following file:

https://www.iana.org/assignments/ipp-registrations

The registry entries will contain the following information:

```
Operation Name                                          Reference
--------------                                          ---------
Check-For-New-Printer-Firmware                          [FWUPDATE]
Update-Printer-Firmware                                 [FWUPDATE]
```

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The authors would also like to thank the following individuals for their contributions to this specification:

Bill Wagner
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15. Change History

This section will be removed when this document is published.

15.1 October 31, 2026

Updated following review in the October 8, 2026 teleconference:

- Changed to Stable status since it has already been prototyped and re-prototyped for this revision;
- Refactored the set of “printer-state-reasons” keywords to align with the set normalized across this specification and IPP System Service [PWG 5100.22];
- Updated IANA Considerations; and
- Various minor editorial changes.

15.2 September 23, 2026

Updated following review at the August 2026 F2F and the September 10, 2026 teleconference, including changes listed on the Firmware Updates wiki page (https://github.com/istopwg/ippsample/wiki/Firmware-Updates).

- Fixed editorial issues in section 4, including updates to Figures 3 and 4;
- Renamed the ‘-failure’ keyword suffixes for “printer-state-reasons” to use ‘-failed’ instead;
- Refactored the set of “printer-state-reasons” keywords to comply with the wiki page guidance, including renaming, deleting, and adding keywords;
- Moved the scheduled update paragraphs of section 5.2 to a new 5.2.1 subsection, and moved the state / “printer-state-reasons” transition guidance paragraphs to a new section 4.6 “Firmware Update”; and
- Updated IANA definitions.

15.3 August 14, 2026

Updated following prototyping the specification in C (libcups fork) and Dart / Flutter (cleanroom):

- Resolved issues identified in the 2026-07-30 IPP WG teleconference review of the start of the previous draft;

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- Added the ‘firmware-update-failure’, ‘firmware-update-in-progress’ and ‘firmware-update-success’ keywords for “printer-state-reasons” to allow the Client to clearly know that a firmware update was in progress or failed or succeeded, rather than having to look for other more detailed state reasons; and

- Updated the definition of the ‘firmware-repository-access-error’ and ‘firmware-repository-unreachable’ keywords to more clearly define which keywords the Printer must remove when each of these keywords is added to “printer-state-reasons”.

15.4 July 29, 2026

Updated with a few minor changes from the July 16 teleconference:

- Removed the “printer-support-uri” attribute from the 2026-07-16 draft, since “printer-more-info-manufacturer” [STD92] is already defined. Added “printer-more-info-manufacturer” to section 11;

- Updated IANA Considerations and section 7.3 to rename ‘xxx-download-xxx’ keywords to be ‘xxx-acquisition-xxx’;

- Added ‘xxx-activation-xxx’ keywords to section 7.3;

- Added a reference to RFC 9019; and

- Updated Section 11 to fix editorial issues.

15.5 July 16, 2026

Updated with changes from teleconference reviews and email feedback

- Changed the operation enum values for the 2 new operations because the ones listed previously were already listed in the draft for TRUSTNOONE;

- Added a new Firmware installation phase state diagram as Figure 1 in section 4;

- Added “-failure”, “-in-progress”, and “-success” keywords for each of the phases in the Figure 1;

- Fixed all the other points from the May F2F;

- Added “printer-firmware-info-uri” attribute, and renamed “printer-new-firmware-support-uri” to be “printer-new-firmware-info-uri”, to provide a URI to release notes, with more accurate attributes’ names;

- Added “printer-support-uri” to specifically point to the printer manufacturer’s model support page;

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- Updated section 11 with implementation recommendations; and
- Update section 10 to add suggested additions from the mailing list.

15.6 May 6, 2026

Updated with some minor fixes identified in the April 23 teleconference, in advance of the May 2026 F2F:

- Added a new “ipp-features-supported” keyword ‘fwupdate’ to indicate support for the Firmware Update feature;
- Updated section 4 to include stages, and revised the text of section 4.3;
- Changed several instances of “Firmware update” to instead be “available new Firmware” and similar editorial fixes; and
- Fixed the IANA Considerations that had some wacky field cross-references that got out of control and otherwise updated to be current with the definitions earlier in the document, added in “printer-state-reasons”, and removed Status Codes section (thanks Mike!).

15.7 April 22, 2026

Updated with fixes to resolve issues identified in reviews from February 26 and March 26:

- Revised names of a number of attributes and their signatures;
- Updated section 4;
- Made many small changes to make things consistent; large scale changes have calmed down; and
- Updated language to normalize terminology for “new Firmware” vs. “Firmware upgrade”.

15.8 February 25, 2026

Updated as per feedback from February 2026 F2F:

- Updated Abstract
- Updated Introduction
- Removed unused definitions in section 2

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- Updated section 4 including Figure 1 and added Table 1, and aligned on how to handle the attributes in Table 1

- Added new “printer-new-firmware-policy-configured” and “printer-new-firmware-policy-supported” attributes, to support section 4.5

15.9 January 27, 2026

Several updates that provide a better structure now that we have consensus on the semantics of the operations

- Updates to section 4 to consolidate discussion of the linkage between the new operations, the new attributes, and the new values for existing attributes

- Updated the abstract

- Fixed the document file name

- Added section 4.5 “Enabling or Disabling the Printer Firmware Update Feature” and defined the “printer-firmware-update-enabled” attribute to allow a Client to have an easy way to detect support for the feature, and to provide the Printer and its Administrator with a way to disable the feature.

15.10 November 14, 2025

A number of updates following the review on 2025-10-30

- Updated section 4 to create sub-sections

- Updates sequence diagram in Figure 2

- Updated 5.1 to correct missing attributes and to change the semantics to have a quick response and have the Client monitor the status via other attributes. This was one of the options in the slides and I chose this one and if we want to change it then we can change it. But we need to review section 4 in depth and that will help us nail down the semantics more quickly

- Updates 5.2 as well

- Removed the server-error-resource-unavailable status code, and gave server-error-firmware-update-in-progress 0x050D in its place.

- Added several more “printer-state-reasons” to handle the Firmware Repository being unreachable and a Firmware update being unavailable, and also noted that at some phases of a Firmware update the IPP Printer will be entirely offline and unable to respond to Get-Printer-Attributes queries.

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15.11 October 29, 2025

- Basically rewrote the entire thing again. Too many changes to name.

15.12 May 6, 2025

Updated with changes discussed at the November 2024 F2F and February 2025 F2F:

- Refactored "printer-firmware-update" to eliminate the 1setOf collection and make all members top-level Printer Status attributes;
- Rewrote the Check-For-Firmware-Update and Update-Printer-Firmware operations and some of the attribute descriptions; and
- Updated the sequence diagram in section 4.

15.13 October 24, 2024

Updated with changes discussed in 2024-10-10 IPP WG teleconference and other offline discussions:

- Refactored diagrams and descriptions of scenarios in section 4;
- Created the new "printer-firmware-update" collection attribute, and moved many of the Printer Status attributes describing available new Firmware update under that so that a list of updates could be provided;
- Added the "printer-firmware-repository" attribute to allow the Firmware Repository to be configured; and
- Added the "printer-firmware-update-uri" operation attribute to allow a Client to specify a URI to a Firmware update file not on the configured Firmware Repository, and added a corresponding use case.

15.14 October 10, 2024

Updated with additional diagrams and several re-arrangements for section 4, which now includes both conceptual scenario description diagrams and sequence diagrams that show IPP operations where used.

15.15 September 11, 2024

Updated with feedback from the 2024-08-29 IPP Workgroup teleconference and other email feedback:

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- Updated the definition of "Firmware" using PWG 5110.1 definition and added reference;
- Changed headings of use cases;
- Moved Successful Firmware update from use cases to exceptions;
- Updated the Model and updated the figures;
- Added the "delay-update-until" and "delay-update-until-time" operation attributes, and changed the name of the operation to "Update-Printer-Firmware" operation;
- Added the Check-For-Firmware-Update operation because having a Get-Printer-Attributes do the check as a side effect of requesting a particular Printer Status attribute seemed like a bad idea; and
- Added additional keywords for the "printer-state-reasons" Printer Status attribute.

15.16 August 26, 2024

Initial revision.

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